Sri Chaitanya ACADEMY | Infinity Learn

Test Date: 13th Apr 2025



JEE FINAL TEST SERIES

JEE Advanced Part Test - 2_P1 [2024 P1]

Grade XII Full Syllabus

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. If the number of ways in which 5 apples, 5 bananas, 5 chickoos and 5 oranges (fruits of the same species are alike) can be distributed equally among five persons so that exactly 2 of them get all 4 identical fruits and each of the remaining persons gets exactly 2 kind of fruits, is n then find the sum of the digits in n…….

A) 8 **B)** 9 **C)** 7 **D)** 6

2. Katrina receives e-mail messages. The table below shows, for a random sample of 40 weekends, the number of e-mail messages received by katrina.

Number of e-mail messages	0	1	2	3	4	5	6	7	8
Number of weekdays	2	3	5	6	11	7	3	2	1

The mean of the number odf email messages received per weekday by Katrina is $\lambda - \frac{1}{\lambda}; (\lambda \geq 1)$. If

$A = \{x : x \text{ is a positive integral divisor of } \lambda\}$ & $B = \{x : x \text{ is a positive integral divisor of } \lambda^2\}$.

Then the sum of digits of $n(A \times B) =$

A) 8 **B)** 9 **C)** 7 **D)** 6

3. Let 'X' and 'Y' be two events such that $P\left(\frac{X}{Y}\right) = \frac{1}{2}$ & $P\left(\frac{Y}{X}\right) = \frac{1}{3}$, then which of the following options is **correct**? ($A\Delta B$ denotes $(A - B) \cup (B - A)$)

A) 'X' and 'Y' can be exclusive events

B) 'X' and 'Y' can be exhaustive events

C) If 'X' and 'Y' are independent events, they can't be exhaustive events

D) $P(X\Delta Y) \leq \frac{3}{4}$

4. F_1, F_2 are left and right focus points of the hyperbola $C: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 (a > 0, b > 0)$. Point O is the origin of the coordinate, M is an arbitrary point on C and above the x-axis. H is a point of MF_1. Given that $MF_2 \perp F_1F_2, MF_1 \perp OH, |OH| = \lambda |OF_2|$, where $\lambda \in \left(\frac{1}{3}, \frac{1}{2}\right)$. Find the range of the eccentricity of the hyperbola C.

A) $(1, \sqrt{3})$ **B)** $(1, \sqrt{2})$ **C)** $(\sqrt{2}, \sqrt{3})$ **D)** $(\sqrt{2}, 2)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. Consider the function $f(x)=\int_0^x\left(5\ell n\left(1+t^2\right)-10t\left(\tan^{-1}t\right)+16\sin t\right)dt$ then which of the following is/are true?

A) $f(x)$ is increasing, $\forall x\in(0,1)$. **B)** $f(x)$ is decreasing, for $x\in(0,1)$.

C) $\int_0^x f(t)dt$ is positive $\forall x\in(0,1)$. **D)** $\int_0^x f(t)dt$ is increasing for $x\in(0,1)$.

6. Let a, b, c be distinct non-zero complex numbers with $|a|=|b|=|c|$ and if each of the equations $az^2+bz+c=0$, $bz^2+cz+a=0$ has a root of modulus 1, then which of the following is/are true

A) $b^2=ac$ **B)** $(a-b)^2+(b-c)^2+(c-a)^2=0$

C) $(a-c)^2=(a-b)(b-c)$ **D)** $|a-b|=|b-c|=|c-a|$

7. Set of all parabola(s) having common chord of maximum length with the ellipse $9x^2+25y^2=225,$ having directrix $x+my+7=0$ $(m\in R)$ then which of the following is/are true?

A) If only one such parabola exists, then $m\in\left\{\frac{-2\sqrt{6}}{5},0,\frac{2\sqrt{6}}{5}\right\}$.

B) If only one such parabola exists, then $m\in\left\{\frac{-2\sqrt{6}}{5},\frac{2\sqrt{6}}{5}\right\}$.

C) If two such parabolas exist, then $m\in\left(\frac{-2\sqrt{6}}{5},\frac{2\sqrt{6}}{5}\right)-\{0\}$

D) If two such parabolas exist, then the range of m *is* $(-3,3)$.

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

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8. Two distinct numbers are selected from the set $\{1,2,3,.....35,36,37\}$ so that the sum of the remaining 35 numbers is the product of these two numbers, then the sum of these two numbers is _____

9. If $\int_a^b \ln\left(\frac{3+3x+x^2}{1+x+x^2}\right)dx = 0$. Then the value of $2|a+b| =$ _____

10. All the faces cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed without looking at them and then randomly drawn two cards simultaneously from the remaining. If the probability that, two cards drawn are both aces, is $\frac{P(^{38}C_{20})}{^{40}C_{20}.^{20}C_2}$, then the value of P is.....

11. Number of ways to choose an ordered pair (a,b) of numbers from the set $\{1,2,3,.....,10\}$ such that $|a-b| \le 5$ _____

12. The value of $\int_{-1}^{1}\left[1008+\frac{[x]}{x^{2014}+1}\right]dx$ _____, where $[.]$ G.I.F

13. Let P_1 and P_2 be two parallelepipeds with non zero volume, formed using the following vectors as their sides respectively.

$P_1 : \left(\hat{i}+\hat{j}+\hat{k}\right), \left(c\hat{i}+a\hat{j}+b\hat{k}\right), \left(b\hat{i}+cj+a\hat{k}\right)$

$P_2 : \left(\hat{i}-\hat{k}\right), \left(3c\hat{i}+a\hat{j}-a\hat{k}\right), \left(b\hat{j}+c\hat{k}\right)$

Where $a,b,c \in \{0,1,2,3,5,6,7,8,9\}$. Let α be the volume of parallelepiped formed by body diagonals of P_1 and β be volume of P_2. Let α,β be related as $\alpha^3+\alpha^2\beta = 4\beta^2\alpha+4\alpha^2\beta$ then the number of unordered triplets (a,b,c) are _____

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14. Consider the equation $f(x) = x^4+2ax^3+(2a-2)x^2+(3-4a)x-2=0$

A fair die is thrown, to select values of "a"

	List-I	List-II	
A)	When $a=0.75$, sum of the absolute value of real roots of the equation $f(x)=0$, is	P)	6
B)	When $a=1.25$, number of real roots of the equation $f(x)=0$, is	Q)	4
C)	Probability that the equation $f(x)=0$ has four real roots is $\frac{m}{n}$, when $2m-n$ is equal to	R)	3

	[m,n are coprime].		
D)	Probability that the equation $f(x)=0$ has exactly two real roots is P, then the largest prime divisor of $\frac{1}{P}$ is	S)	2

A) A – P, B – R, C – S, D – Q **B)** A – R, B – Q, C – S, D – P

C) A – R, B – S, C – Q, D – P **D)** A – P, B – Q, C – S, D –R

15. Consider the differential equation $\left(xy^3(1+\cos x)-y\right)dx+xdy=0$ & $y(\pi)=\frac{\sqrt{\pi}}{\sqrt{2}}$ & solution of the differential equation is $\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+3\pi-\frac{\pi^3}{3}$ where $a,b,c,d\in R$.

List-I		List-II	
A)	Value of a	P)	2
B)	Value of b	Q)	3
C)	Value of c	R)	-2
D)	Value of d	S)	1

A) A – P, B – Q, C – R, D – P **B)** A – P, B – Q, C – P, D – R

C) A – R, B – P, C – Q, D – P **D)** A – P, B – R, C – P, D –S

16. $I_{m.n}=\int(\cos 2x)^m\left(1-\tan^2 x\right)^n dx$ & $I_{0,10}+\alpha.I_{1,10}=\beta\sin 2x\left(1-\tan^2 x\right)^{10}+\gamma I_{0,9}+K$ where K is integration constant & $\alpha,\beta,\gamma\in R$.

	COLUMN-I		COLUMN-II
A)	The value of $\frac{1}{4\alpha}$, is	P)	10
B)	The value of $\frac{1}{4\beta}$ is	Q)	4
C)	Value of $\frac{3\alpha}{\beta}$ is	R)	6
D)	The value of 4γ is	S)	5

A) A – P, B – S, C – Q, D – R **B)** A – P, B – R, C – S, D – Q

C) A – S, B – R, C – P, D – Q **D)** A – S, B – P, C – R, D –Q

17. Let $A=(x+2y+z)^{25}$, $B=(x+2y-z)^{25}$ then

	COLUMN-I		COLUMN-II
P)	The number of distinct terms in A+B equals	**1)**	182
Q)	The number of distinct terms in A-B equals	**2)**	169
R)	The number of distinct terms in AB equals	**3)**	676
S)	The sum of all coefficients in A+B equals	**4)**	$2^{25}\left(2^{25}+1\right)$

A) $P\to(2),Q\to(1),R\to(3),S\to(4)$ **B)** $P\to(1),Q\to(3),R\to(2),S\to(4)$

C) $P\to(1),Q\to(2),R\to(3),S\to(4)$ **D)** $P\to(3),Q\to(1),R\to(2),S\to(4)$

PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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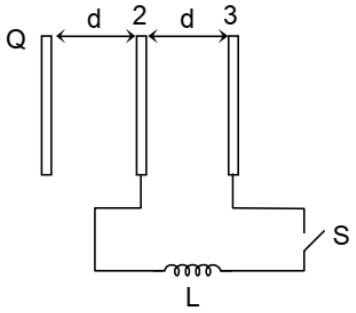
Full Marks : +3 If **ONLY** the correct option is chosen;
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Negative Marks : −1 In all other cases

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18. A thin conducting shell of radius R has surface charge density σ. A small aperture is made is made by removing a circular (disc like) area A from the surface of the disc $\left(A \ll R^2\right)$. The estimated value of charge on the inner surface of shell with aperture is

A) $\frac{\sigma A}{2}$ **B)** $\frac{\sigma A}{4}$ **C)** $\frac{2\sigma A}{3}$ **D)** $\frac{\sigma A}{3}$

19. Three identical large metal plates are fixed at separation of d from each other as shown. The area of each plate is A. Plate 1 is given charge $+Q$ while plates 2 and 3 are neutral and are connected to each other through coil of inductance L and switch S. If resistance of all connected wires is neglected, the maximum current flown through coil after closing switch is $\left(C = \epsilon_0 A / d\right)$ (neglect fringe effect)



A) $\frac{Q_0}{\sqrt{LC}}$ **B)** $\frac{Q_0}{\sqrt{2LC}}$ **C)** $\frac{2Q_0}{\sqrt{LC}}$ **D)** $\frac{Q_0}{2\sqrt{LC}}$

20. A conducting disc of conductivity σ has radius a and thickness δ. A uniform magnetic field $\vec{B}$ is applied in a direction perpendicular to the plane of the disc. The cylindrical region of the magnetic field is coaxial with the disc. If the magnetic field changes with time at the constant rate of $\mathrm{dB/dt}$, then the power dissipated in the disc due to the induced current is

A) $\frac{\pi\delta\sigma a^4}{8}\left(\frac{dB}{dt}\right)^2$ **B)** $\frac{\pi\delta\sigma a^4}{12}\left(\frac{dB}{dt}\right)^2$ **C)** $\frac{\pi\delta\sigma a^4}{4}\left(\frac{dB}{dt}\right)^2$ **D)** $\frac{\pi\delta\sigma a^4}{6}\left(\frac{dB}{dt}\right)^2$

21. A spectrometer is used to observe the Balmer series in the hydrogen spectrum (transitions from quantum number $\mathrm{n} > 2$ to $\mathrm{n} = 2$). The fractional change in wavelength depends upon the quantum number n. The maximum change in wavelength which the spectrometer can measure in the reciprocal of resolving power R. (For change in quantum number $\mathrm{dn} = 1$) take $\left[\mathrm{k} = \frac{13.6}{\mathrm{hc}}\right]$ then

A) $\frac{\mathrm{d}\lambda}{\lambda} = \frac{-2\mathrm{k}\lambda^2}{\mathrm{n}^3}$ **B)** $\frac{\mathrm{d}\lambda}{\lambda} = \frac{-2\mathrm{k}\lambda}{\mathrm{n}^3}$ **C)** $\frac{d\lambda}{\lambda} = \frac{-3\mathrm{k}\lambda^2}{\mathrm{n}^3}$ **D)** $\frac{\mathrm{d}\lambda}{\lambda} = \frac{-3\mathrm{k}\lambda}{\mathrm{n}^3}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

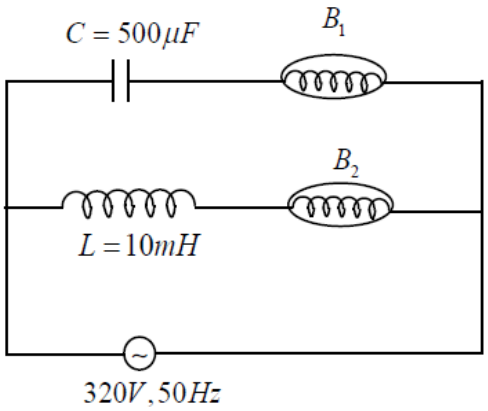
This section contains **THERE (03)** questions.

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22. In the circuit shown in the fig, if both the bulbs B_1 and B_2 are identical and purely resistive then



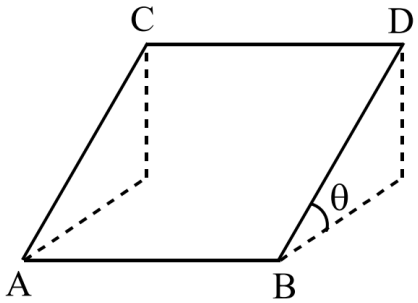
A) Their brightness will be the same

B) B_2 will be brighter than B_1

C) As frequency of supply voltage is increased, brightness of B_1 will increase and that of B_2 will decreases

D) Only B_2 will glow because the capacitor has infinite impedance.

23. According to quantum mechanics, a particle of momentum P can be regarded as a plane matter wave with wavelength $\lambda = \frac{h}{P}$, where h is plank constant. As shown in figure, ABDC is a flat square of side length L at an inclined angle $\theta = 30^\circ$ to the horizontal plane. A neutron beam of kinetic energy E_0 is by some mechanism divided into two beam at point A. One beam moves along path ACD and other beam along the path ABD. Assume no change in speed of neutron while turning at B & C. When the two beam meet at point D they interfere. The mass of neutron is m. Gravitational acceleration is g.



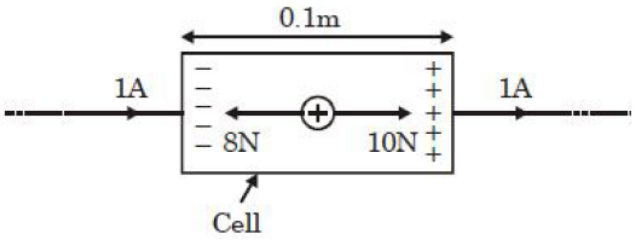
A) At point D matter wave following path ABD leads in phase over other one following path ACD

B) At point D matter wave following path ACD leads in phase over one following path ABD

C) $\sqrt{2mE_0}-\sqrt{m\left(2E_0-mgL\right)}-\frac{h}{2L}=0$; is a condition which satisfies interference minima at D

D) $\sqrt{2mE_0}-\sqrt{m\left(2E_0-mgL\right)}-\frac{h}{2L}=0$; is a condition which satisfies interference maxima at point D

24. The figure shows a conceptualized model of a cell in which unit positive charge experiences a uniform and constant non electrostatic force of (due to chemicals) 10 N and a uniform and constant electrostatic force of 8 N in directions shown in the figure. Find the emf of the cell, the potential difference across the cell and the internal resistance of the cell, and mark the correct option(s).



A) Emf of the cell $=0.2$ V

B) The cell is in charging mode

C) Internal resistance of the cell $=0.2\Omega$

D) Potential difference across the cell $=0.8$ V

SECTION-III

(NON-NEGATIVE INTEGER.)

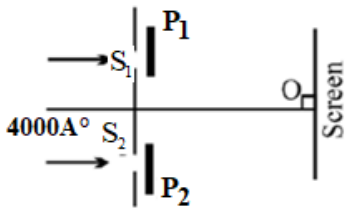
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
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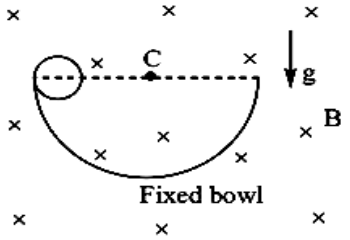
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25. P_1, P_2 are transparent plates having equal thickness 20μ m and refractive indices $\mu_1=1.55, \mu_2=1.5$.. S_1, S_2 are identical slits. The plate P_1 transmits 100% whereas P_2 transmits 25% of energy incident. Without P_1 & P_2, intensity at O is I. the intensity at centre of screen O after placing P_1 and P_2 is $\frac{I}{4N}$. The wave length of incident beam is 4000A°. Then ' N ' is

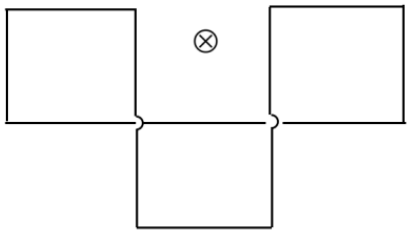


26. A solid sphere is held at rest on the fixed hemispherical bowl made of insulating material such that centre of sphere lies on the horizontal diameter as shown in figure. Total charge $q=5.1\times10^{-1}\text{C}$ and mass $m=42$ grams, both are distributed uniformly in the sphere. There

exists uniform magnetic field B is space as shown in figure. Find limiting value of B (in Tesla) so that on releasing sphere, it does not leaves contact with bowl in course of motion. Consider contact surface to be sufficient rough. Spherical bowl is made of non-magnetic material. Radius of bowl $=10\ \text{cm}$, radius of sphere 3 cm. Take $g=10\ \text{m/s}^2$.

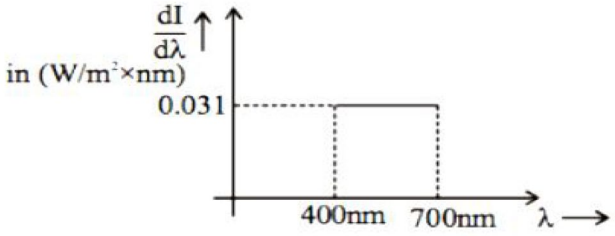


27. Figure shows a conducting loop made of three identical coplanar squares. Side of each square is 1.2 m . Perpendicular to the plane of loop in complete space there exists uniform time dependent magnetic field $B=20t$ Tesla, where t is in sec. Resistance per unit length of wires of each square is $1\Omega/\text{m}$. Find current in loop in amperes?



28. Consider a long cylindrical wire carrying current along the axis of wire. The current distributed non-uniformly on cross section. The magnetic field B inside the wire varies with distance ' r ' from the axis as $B=kr^4$ where k is a constant. The current density J varies with distance ' r ' as $J=cr^n$, then the value of n is (where c is another constant).

29. Scientists have made a light source whose spectral emissive power $\frac{dI}{d\lambda}$ is constant over visible range. Here I is intensity and λ is wavelength. In other words, $\frac{dI}{d\lambda}$ graph is shown below. This beam of area $1\ \text{m}^2$ is incident on emitter plate of photoelectric tube. The collector plate is sufficiently positive so that tube is in saturation mode. Assume that each capable photon liberates 1 electron. If work function is 2 eV , what is the current (in A) in the tube. Round-off the answer to the nearest integer $\left(\lambda(\text{nm})=\left(\frac{1240}{E(\text{eV})}\right)\right)$



30. A point source is kept on the principal axis in front of a convex lens of focal length f, at a distance 4 f from it. Behind the lens, there is a concave mirror of focal length f . There are two positions of mirror, such that final image coincides with the point source. If the ratio of the distance between lens and mirror for the two cases is $\frac{n}{2}$ where $\left(\frac{n}{2}>1\right)$, find the value of n.

SECTION – IV
(MATCHING TYPE)

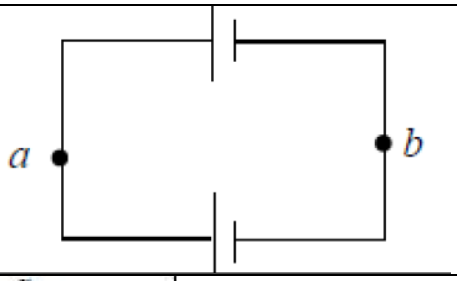
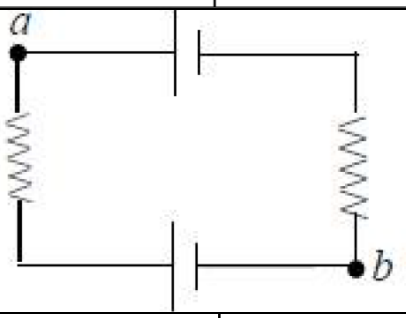
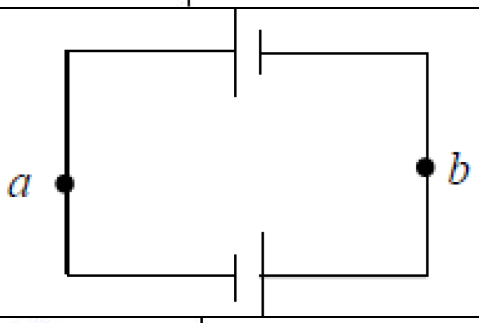
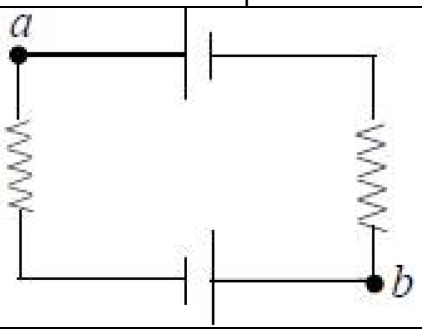
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

---

31. In List I four d.c. circuits are given. In each of them Emf of the cells and their internal resistances are 6 V and 1Ω. In two of the figures the internal resistance is drawn outside the cell. As shown a and b are two points on the circuit V_a and V_b are potential at points a and b.i is current in the circuit. Match List I with List II

	LIST-I		**LIST-II**
I)		**P)**	$V_a - V_b = 0$
II)		**Q)**	$V_a - V_b = 6V$
III)		**R)**	$i = 0$
IV)		**S)**	$i = 6A$
		T)	$i = 3A$

A) I-P,R; II-P,R; III-P,S; IV-P,T **B)** I-Q,R; II-Q,R; III-P,S; IV-P,T

C) I-P,R; II-P,R; III-P,S; IV-P,S **D)** I-Q,R; II-Q,R; III-P,S; IV-P,S

32. LIST - I contains units/dimensions and LIST- II some combination of quantities. Match the LIST - I to List - II.

	LIST-I		LIST-II
I)	Henry	**P)**	$\sqrt{LC}$, where L is inductance and C is capacitance
II)	Second	**Q)**	$\frac{M\ell^2}{Q^2}$, where ℓ is length, M -mass and Q is charge
III)	$(\text{Second})^3$	**R)**	RC, where R is resistance & C is capacitance
IV)	Dimensionless	**S)**	C^2LR, where C is capacitance, $L=$ inductance, R is resistance
		T)	$\frac{Pc}{Q}$, where Q is radiation energy striking per unit area per second. c is speed of light and P is radiation pressure

A) $I-Q, II-P,R; III-S, IV-T$ **B)** $I-T, II-P,R; III-Q, IV-S$

C) $I-T, II-R, III-P,T; IV-Q$ **D)** $I-Q, II-P, III-T, IV-R,S$

33.

	LIST-I		LIST-II
I)	Inductance of a coil	**P)**	Depends on resistivity
II)	Capacitance of a capacitor	**Q)**	Depends on shape
III)	Impedance of a coil	**R)**	Depends on medium inserted
IV)	Reactance of a capacitor	**S)**	Depends on the nature of external voltage source

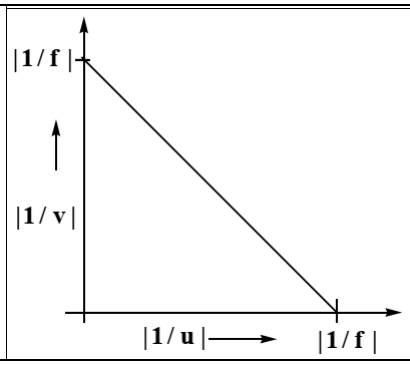
A) I - QR, II - QR, III - PQRS, IV - QRS

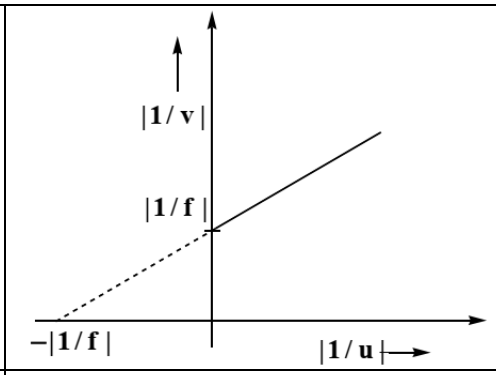
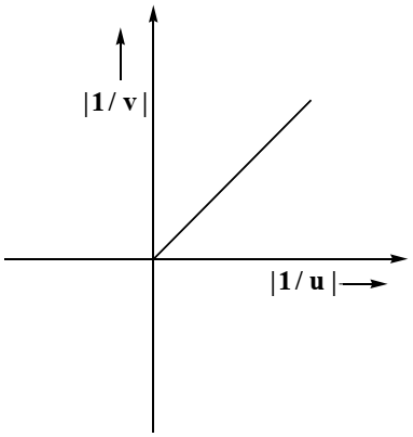
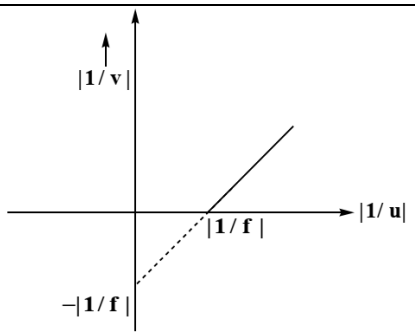
B) I - QS, II - QR, III - PQRS, IV - QRS

C) I - QR, II - QS, III - PQRS, IV - QRS

D) I - QR, II - QR, III - QRS, IV – PQRS

34. Match the graphs of $\left|\frac{1}{V}\right|$ vs $\left|\frac{1}{u}\right|$ where V is image distance and u is object distance for an object which could be real or virtual, under the conditions given below :

	LIST-I		LIST-II
I)	In concave mirror when object is real or virtual and image is real	**P)**	

II)	In concave mirror when object is real and image is virtual	**Q)**	
III)	In convex mirror when object is real or virtual and image is virtual	**R)**	
IV)	In convex mirror when object is virtual and image is real	**S)**	

A) I - P, II - S, III - PQ, IV - R **B)** I - PQ, II - S, III - PQ, IV - S

C) I - PQ, II - S, III - PQ, IV - R **D)** I - PQ, II - S, III - R, IV – PQ

CHEMISTRY **Max Marks: 60**

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

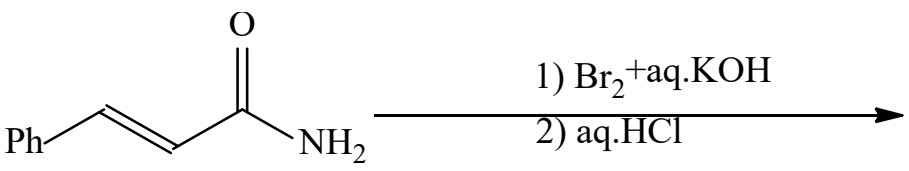
This section contains **FOUR (04)** questions.

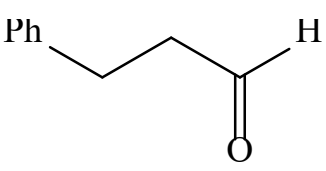
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>

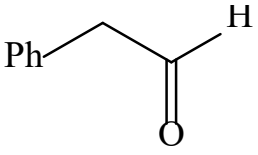
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

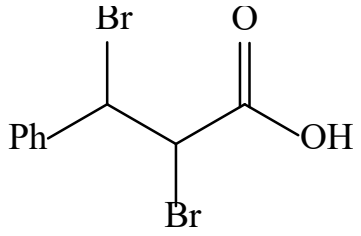
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35. The major product in the following reaction is:

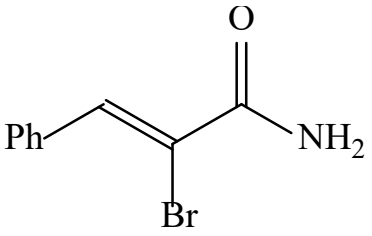


A) 

B) 



C)



D)

36. Two liquids A and B form a non-ideal solution which obey the equation $P_T = P_A^0 + 3\left(P_B^0 - P_A^0\right)x_B + 2\left(P_A^0 - P_B^0\right)x_B{}^2$. When equimolar mixture of A and B is distilled the composition(by mole fraction χ) when this mixture will have a single boiling point is

A) $\chi_A = 0.2$ **B)** $\chi_A = 0.25$ **C)** $\chi_A = 0.4$ **D)** $\chi_A = 0.5$

37. Limiting equivalent conductivity of K^+, Al^{3+} and $SO_4{}^{2-}$ are respectively X, Y and $Z\ S.cm^2eq^{-1}$. Limiting equivalent conductivity of double salt $K_2SO_4.\mathrm{Al}_2\left(SO_4\right)_3.24H_2O$ is (in $S.cm^2eq^{-1}$).

A) $X+Y+Z$ **B)** $2X+2Y+4Z$ **C)** $\frac{X}{4}+\frac{3Y}{4}+Z$ **D)** $4X+\frac{4Y}{3}+Z$

38. Complex X of composition $Cr\left(H_2O\right)_6 Cl_n$ has a spin only magnetic moment of $3.83BM$. It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is

A) Hexa aqua chromium (III) chloride
B) Tetra aqua dichlorido chromium (IV) chloride dihydrate
C) Dichlorido tetra aqua chromium (IV) chloride dihydrate
D) Tetra aqua dichlorido chromium (III) chloride dihydrate

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Among the following suitable route(s) for the conversion of benzaldehyde to acetophenone?

A) CH_3COCl / Anhydrous $AlCl_3$

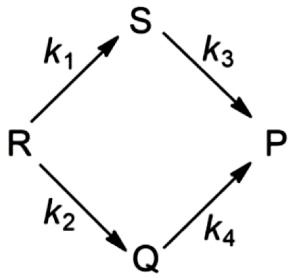
B) I) $\mathrm{HS}-\left(CH_2\right)_3-SH, F_3B.OEt_2$ II) $n-BuLi$ III) MeI IV) $\mathrm{HgCl}_2, CdCO_3, H_2O$

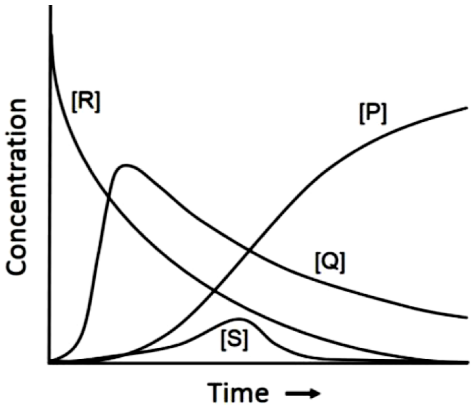
C) $NaNH_2, MeI$

D) I) $MeMgBr$ II) $aq.acid$ III) PCC in CH_2Cl_2

40. For the reaction shown in Scheme 1, the concentration profiles of different species are provided. Based on this graph, the correct condition(s) regarding the rate constants is(are)

Scheme 1





A) $k_2 > k_4$ **B)** $k_3 > k_1$ **C)** $k_2 > k_1$ **D)** $k_1 = k_2$

41. Mn, Fe and Co in the +2 and +3 oxidation states, and all are from hex- aqua complex in acidic aqueous solution. The reduction reactions of the three species are given below where the water ligands are not shown for simplicity. The correct statements about these reactions is / are

$Mn^{3+} + e^- \rightarrow Mn^{2+}$ $\quad E^\circ = +1.57$

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ $\quad E^\circ = +0.77$

$Co^{3+} + e^- \rightarrow Co^{2+}$ $\quad E^\circ = 1.97$

A) Mn^{3+} and Co^{3+} are more easily reduced than Fe^{3+}

B) $Fe^{3+}(aq)$ is stable due to high spin stable $t_{2g}^3 e_g^2$ configuration (Jahn Teller distortion is not observed).

C) Reduction of $Mn^{3+}(aq)$ and $Co^{3+}(aq)$ to the corresponding stable $Mn^{2+}(aq) Co^{2+}$ can takes place

D) Fe^{3+} is a weaker oxidizing agent compared to Mn^{3+} and Co^{3+} in aqueous solution

SECTION-III

(NON-NEGATIVE INTEGER.)

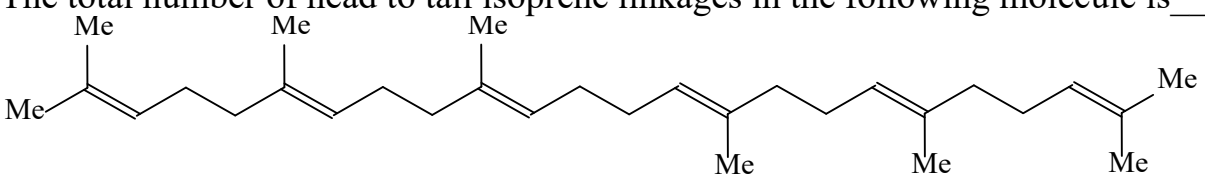
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

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42. The total number of head to tail isoprene linkages in the following molecule is_____



43. Gram molecular weight (in g) of sucralose is _____

[Given, atomic weight: $H = 1, C = 12, N = 14, O = 16, Na = 23, Cl = 35, Ca = 40$]

44. Nitrogen gas adsorbed on charcoal to the extent of $0.387\, cm^3 / g$ at a pressure of 1.6 atm and at temperature of $200K$, but at $250K$ the same amount of adsorption was achieved

only when the pressure was increased to 32 atm. The magnitude of molar enthalpy of adsorption (in kcal/mol) of nitrogen on charcoal is_____ $(\ln 20 = 3.0)$

45. The formation constant of $\left[Co(CN)_6\right]^{4-}$ is 10^{+19}. Formation constant of $\left[Co(CN)_6\right]^{3-}$ in the correct scientific notation is $P \times 10^{+Q}$. $(P+Q)$ value is

$$\left(\text{Given}: E^{\circ}_{Co^{3+}/Co^{2+}} = +1.82V, E_{\left[Co(CN)_6\right]^{3-}/\left[Co(CN)_6\right]^{4-}} = -0.838V, \frac{2.303RT}{F} = 0.06, \log 2 = 0.3\right)$$

46. How many of the following molecules combine with chlorine ?
$CO, SO_2, NO, NO_2, NH_3, PCl_3, FeCl_2, CuCl$

47. For the process

$$\left[Co(NH_3)_5 Cl\right]^{2+} + Cl^- \rightarrow \left[Co(NH_3)_4 Cl_2\right]^{+} + NH_3$$

The expected ratio of cis to trans isomers in the product is (neglect the energy considerations)

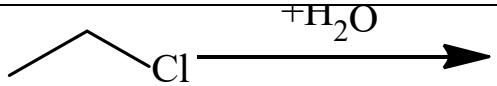
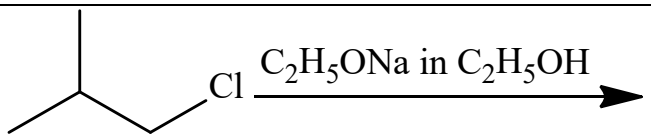
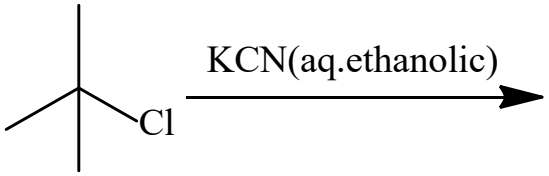
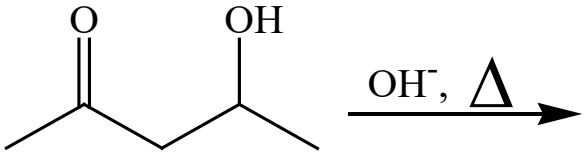
SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

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Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

48.

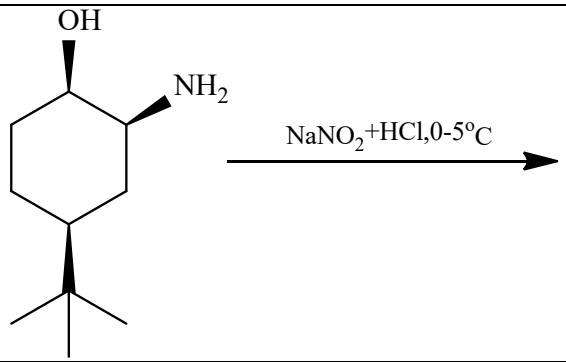
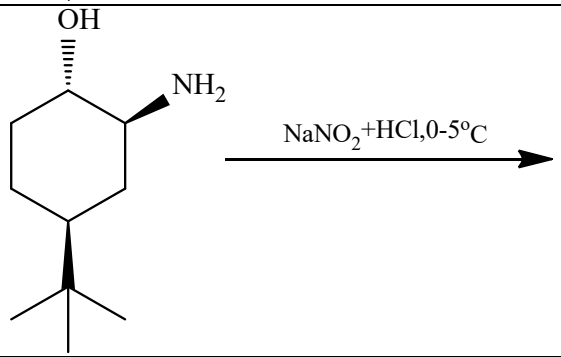
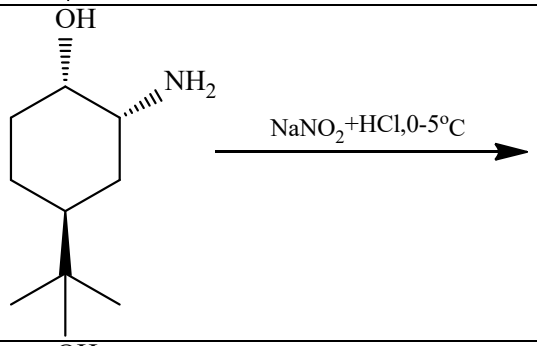
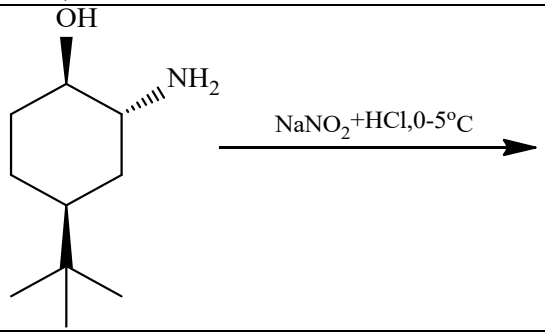
List-I (Reaction)		List-II (Mechanism for the formation of major products)	
P)		**1)**	$S_N{}^1$
Q)		**2)**	$S_N{}^2$
R)		**3)**	E_1
S)		**4)**	E_2
		5)	E_{1CB}
		6)	No reaction

The correct match is

A) $P-6; Q-4; R-2; S-5$ **B)** $P-6; Q-4; R-1; S-5$

C) $P-1; Q-4; R-2; S-5$ **D)** $P-1; Q-3; R-2; S-4$

49.

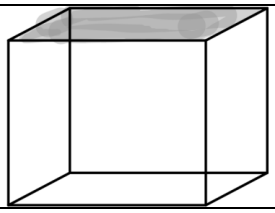
List-I (Reaction)		List-II (Regarding major product)	
P)		**1)**	Gives colored product with $2,4-D.N.P$
Q)		**2)**	Can reduce Tollens reagent
R)		**3)**	Gives $2^{\circ}-$alcohol on reaction with CH_3MgBr followed by acidification.
S)		**4)**	Gives $3^{\circ}-$alcohol on reaction with CH_3MgBr followed by acidification.

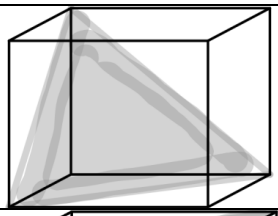
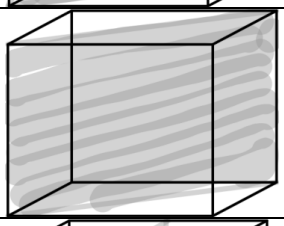
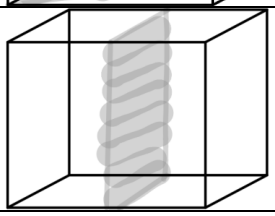
The correct match is

A) $P-1,2,3;Q-1,2,3;R-1,4;S-3$ **B)** $P-1,2,3;Q-1,4;R-1,4;S-1,4$

C) $P-1,4;Q-1,2,3;R-1,4;S-1,2,3$ **D)** $P-1,4;Q-1,4;R-1,4;S-3$

50. In the list-I different planes in F.C.C unit cell given, in the list-II type of voids present in the plane given.

List-I		List-II	
P)		**1)**	Only tetrahedral voids

Q)		**2)**	Only octahedral voids
R)		**3)**	Both tetrahedral and octahedral voids
S)		**4)**	Neither tetrahedral nor octahedral voids

The correct match is

A) $P-2;Q-1;R-3;S-4$ **B)** $P-2;Q-1;R-1;S-2$

C) $P-2;Q-1;R-3;S-2$ **D)** $P-2;Q-4;R-3;S-2$

51. Match the following Column-I with Column-II

	Column I		**Column II**
A)	$FeCl_3(aq)+H_2S(g)\rightarrow$	P)	Colourless solution is obtained
B)	$SnCl_2(aq)+NaOH(excess)\rightarrow$	Q)	White precipitate is formed
C)	$BiCl_3(s)+H_2O(l)\rightarrow$	R)	Green coloured solution is obtained
D)	$CaCl_2(s)+2NaHCO_3\underset{(aq)}{\xrightarrow{\Delta}}$	S)	Non-redox reaction
		T)	Colourless gas is evolved

A) A-QR, B – PS, C – QS, D – PQST **B)** A- PS, B – QS, C – QR, D – QS

C) A- QS, B – QR, C – PS, D – QS **D)** A- PS, B – QS, C – QS, D –QR